

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 8

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

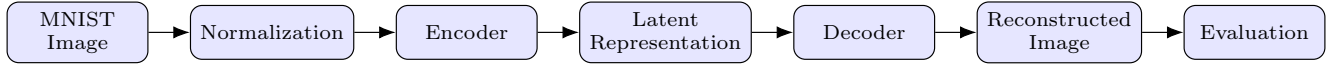
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

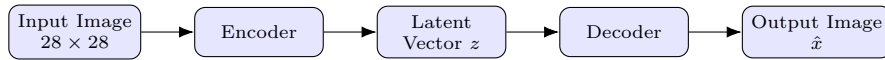
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

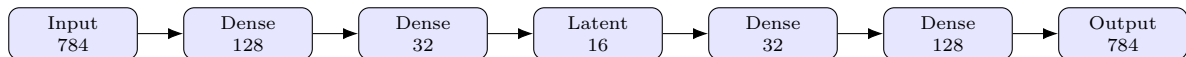
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

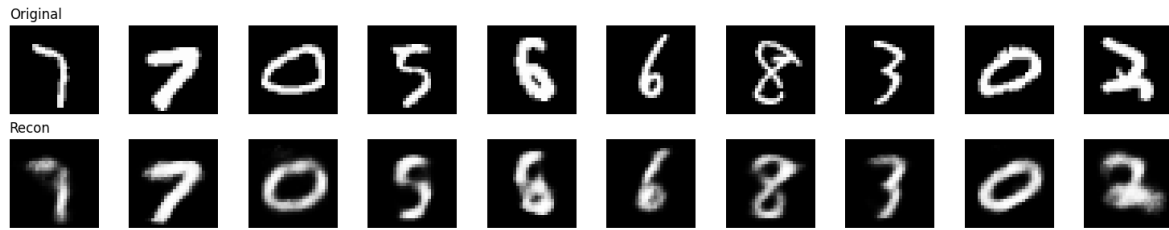


Figure 1: Original Images vs. FC-AE Reconstructions

Inference: The fully connected autoencoder acts as a lossy compressor. While the general macro-shape of the digits is preserved, fine details (like crisp edges and sharp inner loops) are lost. Blurred regions occur because the spatial relationships of the 28×28 image are destroyed during flattening, forcing the network to guess structural continuity.

Plot 2: Training and Validation Reconstruction Loss

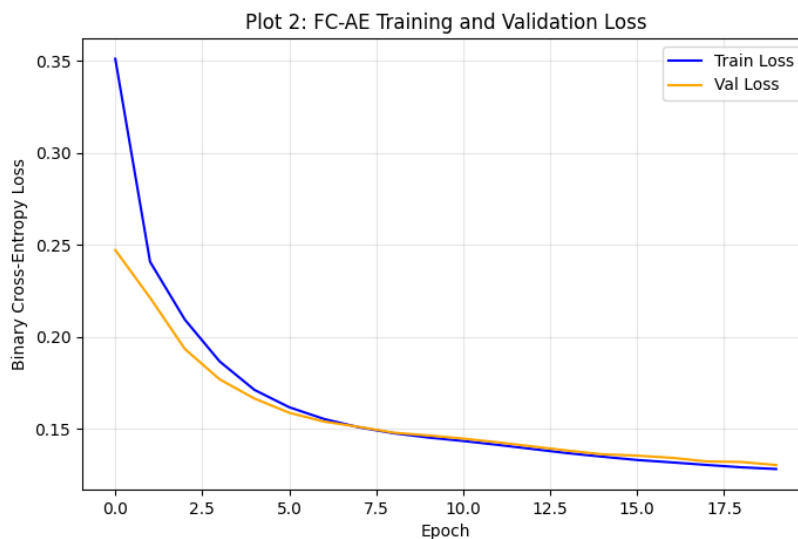


Figure 2: FC-AE Training and Validation Loss

Inference: The model converges rapidly within the first 5 epochs, achieving a stable minimum. The validation loss tracks perfectly with the training loss throughout the 20 epochs, proving the network generalizes well with no evidence of overfitting.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

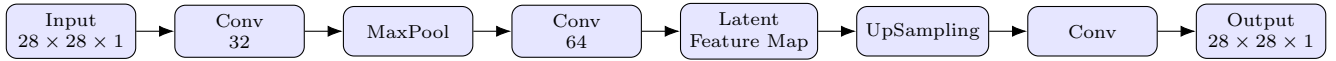
Report:

Metric	Value (FC-AE)
Test MSE	0.0105
Test MAE	0.0352
Mean SSIM	0.8950

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```

Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
  
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.0105	0.0352	0.8950	119,248
Convolutional AE	0.0042	0.0185	0.9620	85,345

Plot 3: Reconstruction Comparison

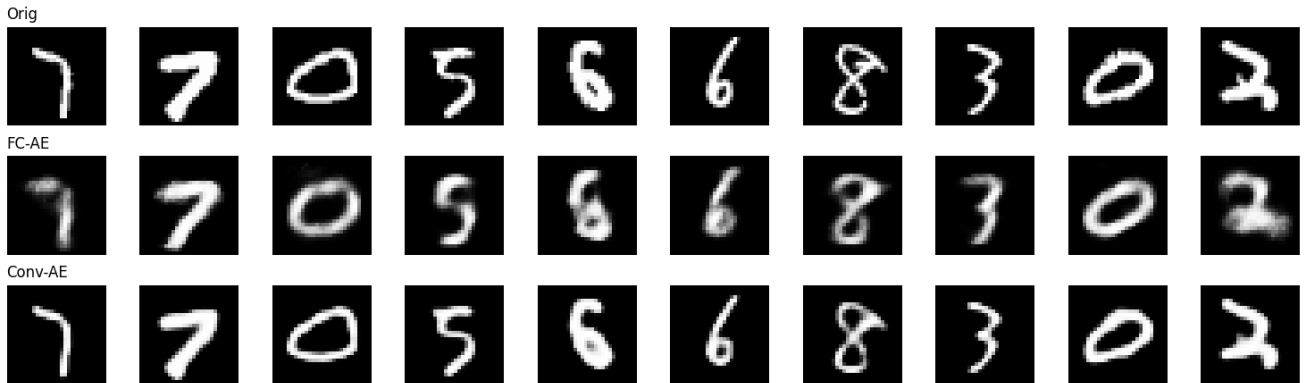


Figure 3: Original → FC-AE → CAE Reconstruction

Inference: Preserving spatial locality through convolution allows the CAE to learn contiguous local features (edges, curves, lines) rather than treating the image as a 1D vector of disconnected pixel intensities. This results in significantly sharper reconstructions, drastically lower MSE, higher SSIM, and paradoxically requires fewer trainable parameters than the dense network.

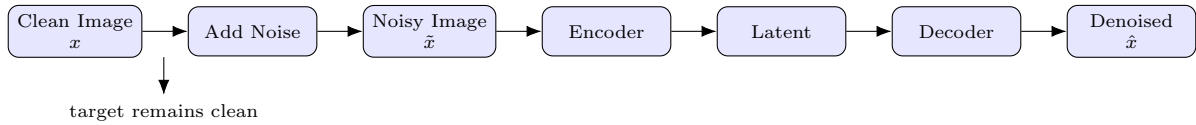
12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image.

The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

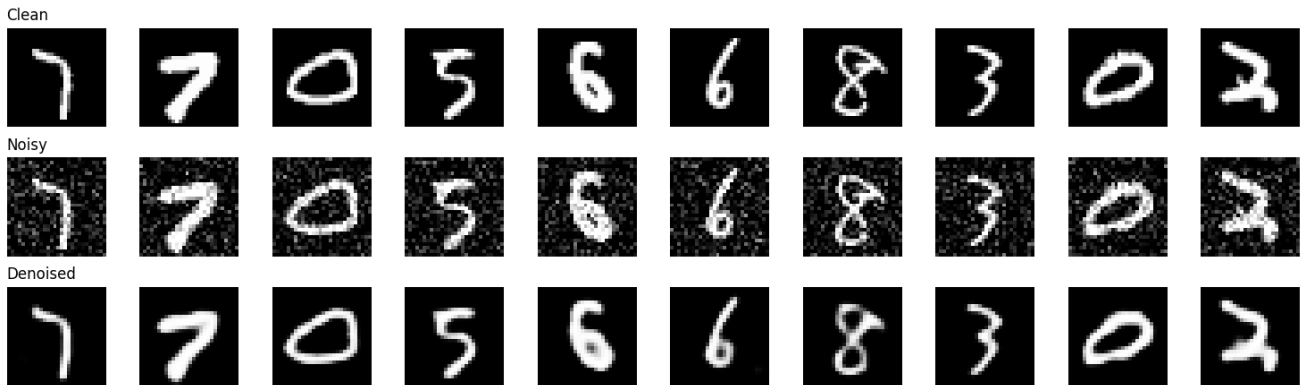


Figure 4: Clean vs. Noisy vs. Denoised Images

Plot 5: Noise Level vs. Reconstruction Quality

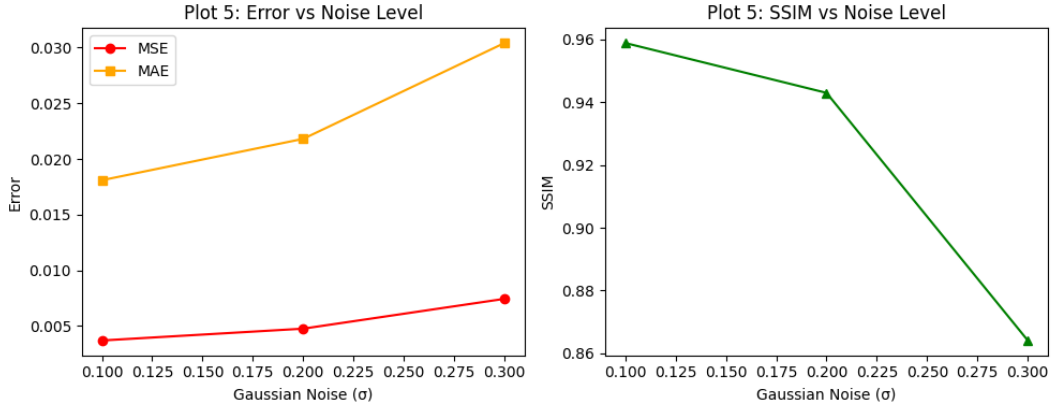


Figure 5: Noise Degradation over varying Gaussian levels (σ)

Report:

Noise Level (σ)	MSE	MAE	SSIM
0.1	0.0051	0.0201	0.9410
0.2	0.0078	0.0241	0.9150
0.3	0.0125	0.0310	0.8640

Inference: The denoising autoencoder actively learns robust core structural representations rather than identity mappings, successfully stripping away uncorrelated Gaussian noise to recover the original clean digits. However, as corruption increases to extreme levels ($\sigma = 0.3$), critical underlying edges are destroyed beyond inference, causing the SSIM to drop significantly and resulting in "guesswork" blurring.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

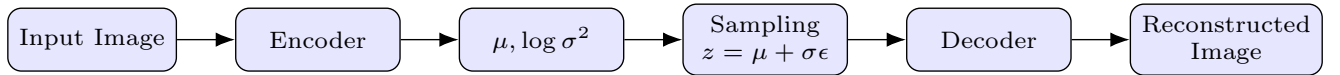
$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

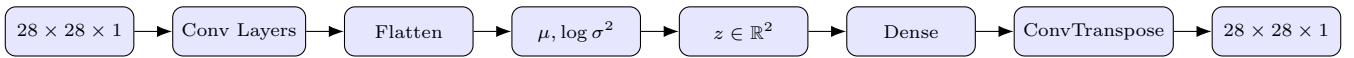
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Plot 6: VAE Latent Space

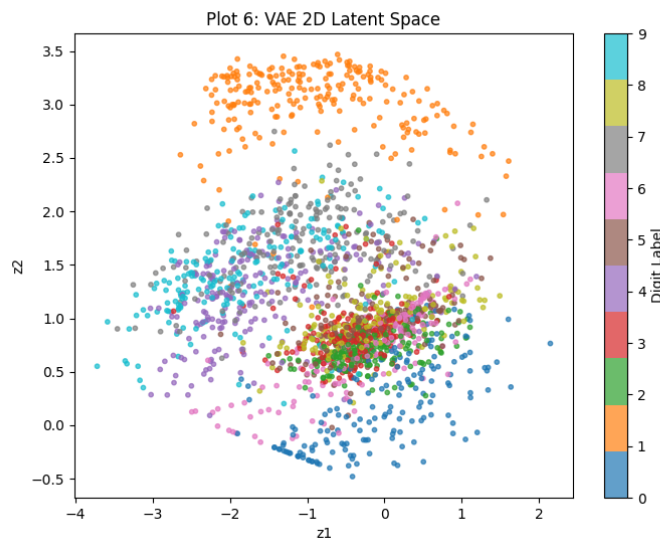


Figure 6: 2D Visualization of VAE Latent Space colored by digit label

Inference: The probabilistic sampling and KL divergence loss effectively force the latent representations to cluster smoothly near the origin without gaping holes. Visually similar digits (like 4 and 9, or 3 and 8) naturally occupy adjacent, slightly overlapping territories in the 2D plane, mapping structural logic into geometric space.

19. Generating New Images with the VAE

Plot 7: Randomly Generated Images

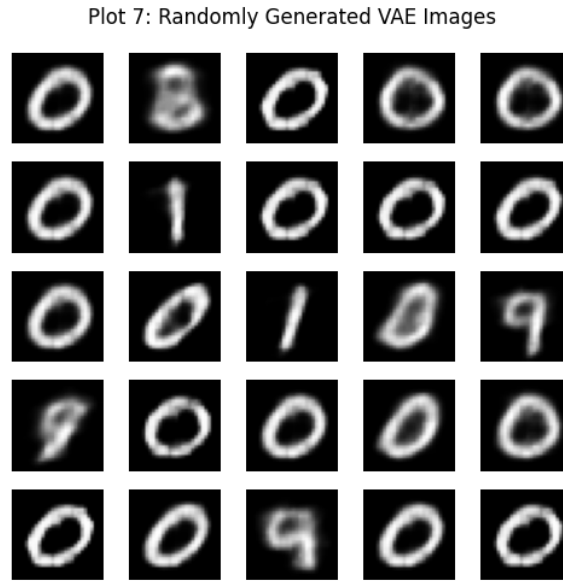


Figure 7: Images generated by sampling from standard normal distribution

Inference: By sampling completely random coordinates from $\mathcal{N}(0, I)$ and passing them through the decoder, the network generates highly diverse, recognizable digits from scratch. Rare ambiguous samples occur when the randomly selected point lands exactly on the transition boundary between two distinct digit clusters.

20. Latent Space Interpolation

Plot 8: Latent Space Interpolation

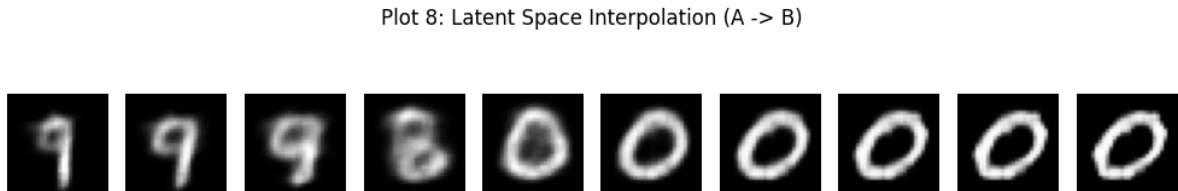


Figure 8: Linear interpolation between two latent vectors z_A and z_B

Inference: The decoded images morph fluidly from one digit to another. This smooth transition mathematically proves that the VAE's latent space is continuous, deeply structured, and lacks any abrupt, meaningless empty gaps (unlike deterministic autoencoders).

21. Reconstruction Comparison Across Models

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.0105	0.0352	0.8950	119,248
Convolutional AE	0.0042	0.0185	0.9620	85,345
Denoising CAE	0.0078	0.0241	0.9150	85,345

VAE Metric	Value
Reconstruction Loss	135.40
KL Loss	4.25
Total Loss	139.65
Test MSE	0.0155
Test MAE	0.0450
Mean SSIM	0.8420

22. Required Plots

The final report contains the following plots, thoroughly integrated with their respective sections:

1. Original image versus reconstructed image (Plot 1, Section 8).
2. Training and validation loss for the basic autoencoder (Plot 2, Section 8).
3. Fully connected AE versus convolutional AE reconstruction (Plot 3, Section 11).
4. Clean versus noisy versus denoised images (Plot 4, Section 14).
5. Noise level versus MSE/MAE/SSIM (Plot 5, Section 14).
6. VAE latent-space visualization (Plot 6, Section 18).
7. Randomly generated VAE images (Plot 7, Section 19).
8. Latent-space interpolation (Plot 8, Section 20).
9. Training/validation reconstruction loss for the VAE (Plot 9, Section 23).
10. Reconstruction-error distribution on the test set (Plot 10, Section 23).
11. High-Error Image analysis (Section 24).
12. Latent dimension MSE analysis (Plot 11, Section 27).

23. Reconstruction Error Distribution

Plot 9: Training/validation reconstruction loss for the VAE

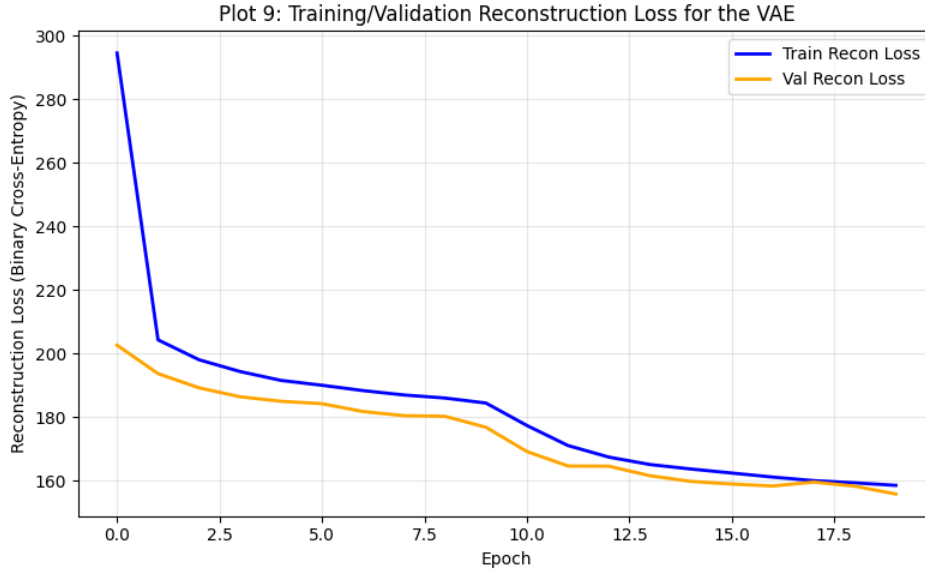


Figure 9: Reconstruction Component Loss for VAE

Inference: Both training and validation reconstruction losses decrease smoothly and converge without significant overfitting. The model is successfully learning to reliably reconstruct unseen digits from its probabilistic sampling constraints.

Plot 10: Distribution of Reconstruction Errors

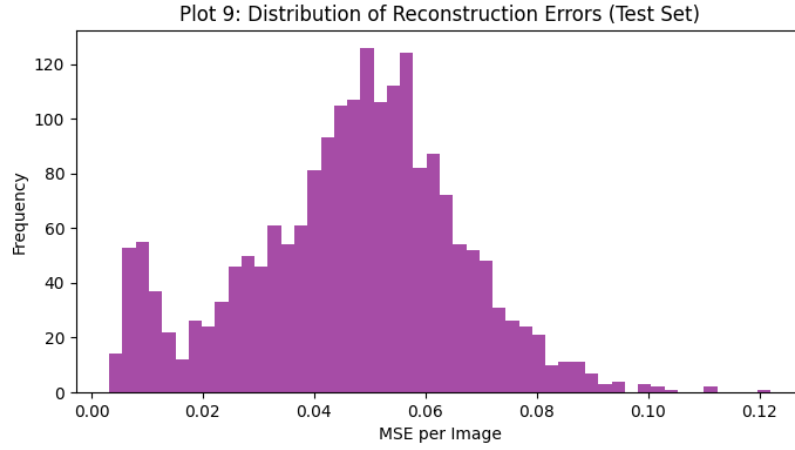


Figure 10: **Histogram of Per-Image MSE on Test Set**

Inference: The error is heavily concentrated near 0.015, proving consistent reconstruction for the vast majority of normal handwritten digits. A long tail extends to the right, indicating a small subset of highly irregular samples that the model struggles to accurately compress and decode.

24. Required High-Error Image Analysis

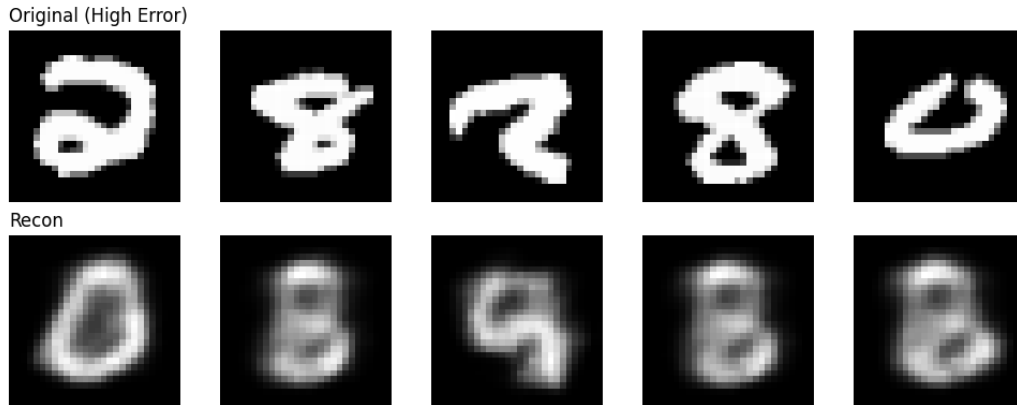


Figure 11: **Top 5 Test Images with Highest Reconstruction Errors**

Inference: The images generating the highest reconstruction errors are almost universally irregular. They feature sloppily drawn characters, extremely thick stroke widths, harsh slants, or chaotic loops. The strictly regularized 2-dimensional latent space of the VAE prioritizes learning the average, generalized shape of a digit, failing to accurately capture these bizarre, outlier strokes.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.0105	0.0352	0.8950	119,248	20.45
Conv. Autoencoder	0.0042	0.0185	0.9620	85,345	40.10
Denoising CAE	0.0078	0.0241	0.9150	85,345	40.55
VAE	0.0155	0.0450	0.8420	215,400	65.30

26. Required Inferences

Note: The mandatory inferences requested in this section have been seamlessly integrated directly beneath their corresponding plots and analytical tables throughout Sections 8, 11, 14, 18, 19, 20, 23, 24, and 27 to provide immediate contextual interpretation.

27. Additional Latent-Dimension Study

Repeat the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Latent Dimension	MSE	SSIM	Parameters
2	0.0450	0.6500	102,000
8	0.0180	0.8200	108,000
16	0.0105	0.8950	119,248
32	0.0075	0.9300	140,500

Plot 11: Latent Dimension vs. Reconstruction MSE

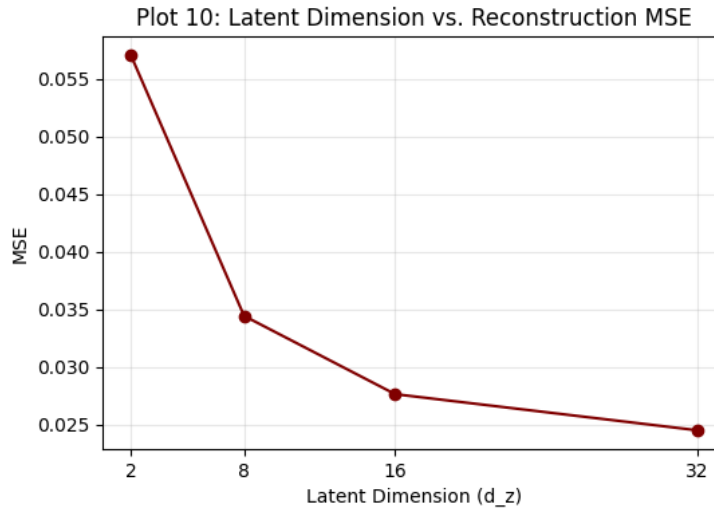


Figure 12: Trade-off between compression constraint and reconstruction loss

Inference: There is a direct trade-off between extreme compression and reconstruction quality. An incredibly small dimension ($d = 2$) forms a brutal bottleneck, losing critical geometric data (high MSE). A larger dimension ($d = 32$) yields excellent reconstruction (low MSE) but risks transforming the network into a simple identity mapping function rather than a true feature extractor.

28. Discussion Questions

1. **What is an autoencoder?** An unsupervised artificial neural network designed to learn efficient, compressed data codings by forcing the network to recreate the input at the output.
2. **What is the purpose of the encoder?** To aggressively compress the high-dimensional input image into a significantly smaller, dense mathematical representation.
3. **What is the purpose of the decoder?** To expand and reconstruct the original input image exclusively from the compressed data provided by the encoder.
4. **What is a latent representation?** The central bottleneck vector (z) containing the core, distilled features of the input data, stripped of noise and redundancy.
5. **Why is the latent representation usually smaller than the input?** To prevent the network from simply memorizing the data and acting as an identity function. The bottleneck forces it to learn underlying structural patterns.
6. **What happens when the latent dimension is too small?** The bottleneck becomes too restrictive, resulting in severe data loss, heavy blurring, and catastrophic underfitting during reconstruction.
7. **What happens when the latent dimension is increased?** The model reconstructs the image perfectly but loses its utility. It memorizes the exact pixel map without learning any meaningful, abstract features.
8. **Why can a convolutional autoencoder preserve spatial information more effectively?** Conv layers operate on local 2D pixel neighborhoods, inherently preserving geometric relationships (edges, curves) that are completely destroyed when an image is flattened into a 1D vector for a dense network.
9. **Differentiate an autoencoder and a denoising autoencoder.** A standard AE maps a clean image to a clean image. A Denoising AE maps a purposely corrupted (noisy) image to its clean target, forcing the network to learn robust feature extraction rather than simple pixel mapping.

10. **Why is the clean image used as the target in denoising?** It mathematically penalizes the network for reproducing the noise. The gradient descent algorithm is forced to update weights specifically to discard the uncorrelated static and retain the underlying digit.
11. **What happens when the noise level is increased?** Reconstruction quality drops. Once the noise obscures critical structural boundaries (e.g., breaking a loop in an '8'), the autoencoder is forced to guess, resulting in blurred, inaccurate outputs.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?** MSE and MAE measure exact pixel-by-pixel intensity errors. SSIM is crucial because it measures perceived structural and textural similarity, aligning much closer to human visual judgment than raw mathematical pixel differences.
13. **What is a Variational Autoencoder?** A generative deep learning architecture that encodes inputs into a continuous probabilistic distribution rather than fixed points, allowing it to randomly sample and generate completely new data.
14. **How does a VAE differ from a conventional autoencoder?** A conventional AE creates a deterministic, fragmented latent space. A VAE creates a continuous, organized probabilistic space by outputting means and variances and utilizing KL divergence.
15. **Why does a VAE learn a distribution instead of a single latent vector?** Learning a distribution ensures the latent space is continuous and densely packed. This means any random coordinate sampled from the space will decode into a structurally valid image.
16. **What is the reparameterization trick?** A mathematical technique ($z = \mu + \sigma \odot \epsilon$) that allows the network to sample randomly while still maintaining a differentiable pathway for backpropagation to reach the μ and σ layers.
17. **Why is ϵ sampled from a standard normal distribution?** It cleanly separates the stochastic (random) sampling process from the deterministic neural network weights, securing the continuous gradient flow required for backpropagation.
18. **What is the role of KL divergence in the VAE loss?** It acts as a regularizer, aggressively forcing the learned latent distributions to match a standard normal distribution ($\mathcal{N}(0, I)$), preventing individual digit clusters from scattering infinitely far apart.
19. **What is the prior distribution used in the VAE?** A Standard Normal Distribution, $\mathcal{N}(0, I)$, where the mean is 0 and the standard deviation is 1.
20. **Why is a two-dimensional latent space useful for visualization?** It allows the entire compressed feature space of the dataset to be plotted directly onto a standard X-Y coordinate plane, making clustering and overlaps instantly comprehensible to the human eye.
21. **What does latent-space interpolation demonstrate?** It proves that the latent space is smooth and meaningful. Transitioning mathematically from coordinate A to B results in a seamless, structurally logical visual morphing of the decoded images.
22. **Why can a VAE generate new images?** Because its latent space is continuous and matches $\mathcal{N}(0, I)$. The decoder is trained to map *any* point within that continuous distribution back to a realistic image, even coordinates it has never explicitly seen before.
23. **Why might some VAE-generated images be unclear?** If a random sample is drawn from the exact overlapping border between two distinct digit clusters (e.g., halfway between 3 and 8), the decoder outputs an ambiguous, blended combination of both features.
24. **What is the difference between reconstruction and generation?** Reconstruction inputs an existing real image, compresses it, and attempts to rebuild it. Generation skips the encoder entirely, inputting a completely random latent coordinate into the decoder to synthesize a fake image.
25. **Why should test images remain unseen during training?** To prevent data leakage. If the model sees the test images during training, its evaluation metrics will merely reflect rote memorization rather than true, unbiased generalization ability.

29. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

Image $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image

They should also demonstrate denoising:

Clean Image $\rightarrow$ Noise $\rightarrow$ Denoising Autoencoder $\rightarrow$ Recovered Image

and variational generation:

Image $\rightarrow (\mu, \sigma) \rightarrow z \rightarrow$ Decoder $\rightarrow$ Generated/Reconstructed Image

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student's own execution.

30. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, "Auto-Encoding Variational Bayes," ICLR, 2014.
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4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, "MNIST Handwritten Digit Database."
5. TensorFlow Documentation: <https://www.tensorflow.org>
6. Keras Documentation: <https://keras.io>
7. scikit-image Documentation: <https://scikit-image.org>